

Formative Assessment 3

DSC1105 Exploratory Data Analysis

Sindayen, Lorenzo Danilo V.

2026-02-21

GitHub Link:

<https://github.com/ChewyGnome/DSC1105>

Dataset

The dataset that will be used in this report is the 2023 Yellow Taxi Trip Records provided by the NYC Taxi & Limousine Commission (TLC). In this dataset, we have the data regarding the millions of yellow taxi trips throughout the year of 2023. Containing specific details such as the drop off times, trip distance, total fare amount, and etc. The objective of this study is to provide a model function that will calculate the accurate conversion for fare fee to the amount of distance travelled; preferably also taking into account the seasonal differences across fares.

Datasource: <https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page>

```
taxi <- read_parquet("Data/2023_yellow/yellow_tripdata_2023-01.parquet")

taxi_clean <- taxi %>%
  filter(trip_distance > 0,
         total_amount > 0)
```

Part 1: Data Import & Preparation

Data Import & Cleaning

```
taxi_clean <- taxi_clean
```

Variable Derivation

```

taxi_clean <- taxi_clean %>%
  mutate(
    trip_duration_minutes = as.numeric(difftime(tpep_dropoff_datetime,
                                                  tpep_pickup_datetime,
                                                  units = "mins")),
    fare_per_mile = total_amount / trip_distance,
    month = factor(month(tpep_pickup_datetime), levels = 1:12, labels = month.abb)
  )

```

Output:

Summary Table of Variables

```

set.seed(123)
taxi_sample <- taxi_clean %>% sample_n(100000)

taxi_sample %>%
  select(trip_distance, total_amount,
         trip_duration_minutes, fare_per_mile) %>%
  summary() %>%
  knitr::kable(digits = 2)

```

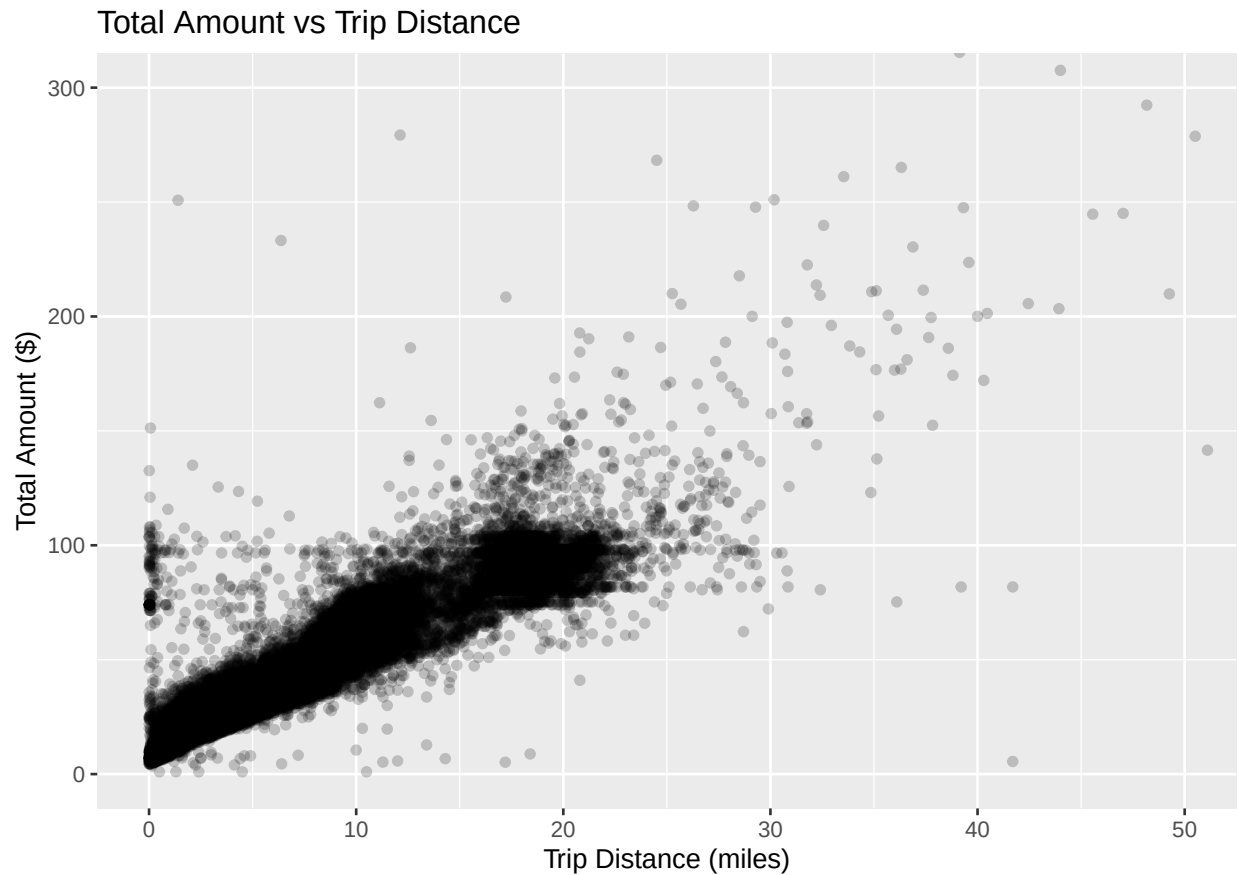
trip_distance	total_amount	trip_duration_minutes	fare_per_mile
Min. : 0.010	Min. : 1.0	Min. : 0.00	Min. :3.800e-03
1st Qu.: 1.100	1st Qu.: 15.6	1st Qu.: 7.25	1st Qu.:7.978e+00
Median : 1.810	Median : 20.2	Median : 11.67	Median :1.087e+01
Mean : 3.504	Mean : 27.4	Mean : 15.95	Mean :1.591e+01
3rd Qu.: 3.400	3rd Qu.: 28.9	3rd Qu.: 18.48	3rd Qu.:1.467e+01
Max. :4223.150	Max. :487.4	Max. :1439.13	Max. :1.326e+04

Scatterplot: Total Fare Amount vs Trip Distance

```

ggplot(taxi_sample, aes(x = trip_distance, y = total_amount)) +
  geom_point(alpha = 0.2) +
  labs(title = "Total Amount vs Trip Distance",
       x = "Trip Distance (miles)",
       y = "Total Amount ($)") +
  coord_cartesian(xlim = c(0, 50), ylim = c(0, 300))

```



Correlation Coefficient

```
cor(taxi_sample$trip_distance,  
    taxi_sample$total_amount,  
    use = "complete.obs") %>%  
  round(4) %>%  
  cat("Correlation coefficient:", .)
```

```
## Correlation coefficient: 0.2413
```

Interpretation: The correlation coefficient of the model indicates that there is a significant correlation between the trip distance and the fare fee for the ride. This shows that longer trips tend to scale a lot better compared to shorter trips because of the initial base price that each trip is given.

Part 2: Classical Linear Regression

Model Fit

```
lm_model <- lm(total_amount ~ trip_distance, data = taxi_sample)

tidy(lm_model) %>% knitr::kable(digits = 3)
```

term	estimate	std.error	statistic	p.value
(Intercept)	26.357	0.067	396.305	0
trip_distance	0.297	0.004	78.618	0

```
glance(lm_model) %>%
  select(r.squared) %>%
  knitr::kable(digits = 4)
```

r.squared
0.0582

Interpretation:

The fitted model is:

$$\text{total_amount} = \beta_0 + \beta_1 \times \text{trip_distance} + \varepsilon$$

Slope (β_1):

For each mile travelled, the fare cost also increases in proportion to the distance increase

Intercept (β_0):

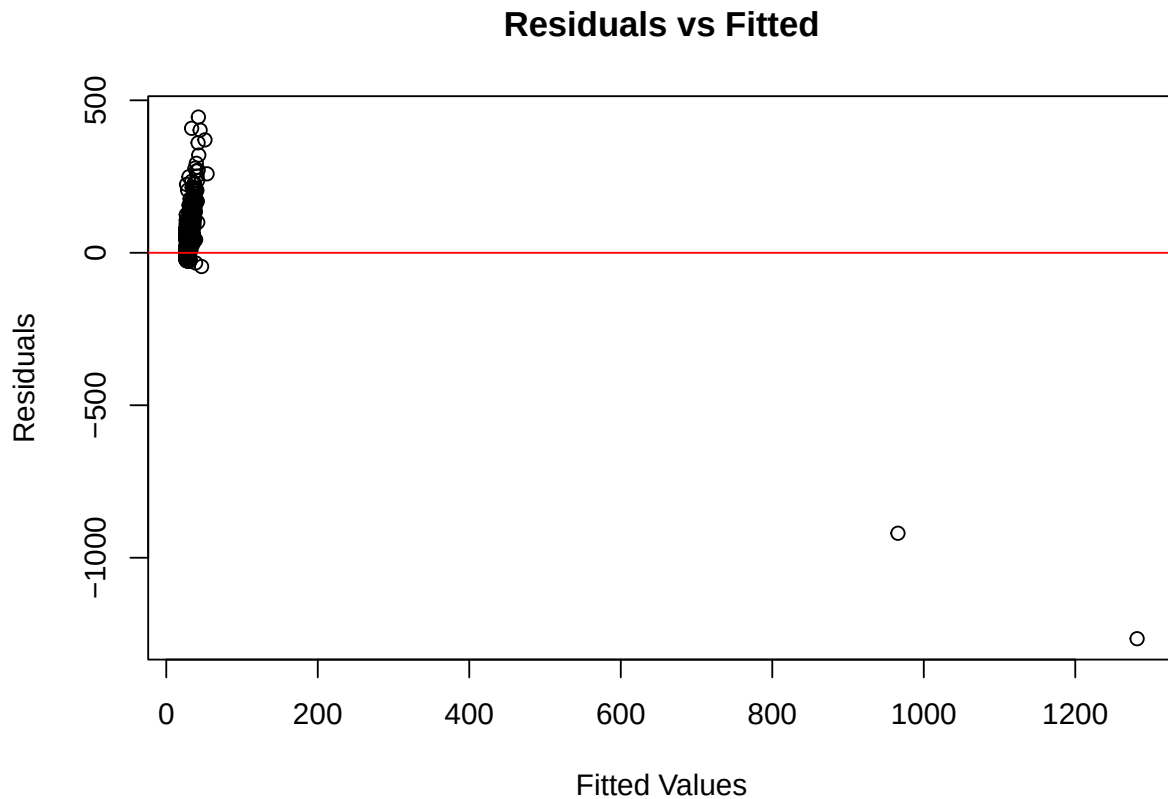
Represents the base fare when distance is 0. While R^2 indicates that trip distance shows a substantial portion of variability in the total fare.

R^2 :

The R^2 value suggest that there is a proportional increase between total amount and trip distance. With having higher yields of R^2 value show a stronger explanatory power of the linear model.

Residuals vs Fitted Plot:

```
plot(lm_model$fitted.values,
     lm_model$residuals,
     xlab = "Fitted Values",
     ylab = "Residuals",
     main = "Residuals vs Fitted")
abline(h = 0, col = "red")
```



Diagnostics Comments:

Linearity:

If the residuals are randomly scattered around zero without a clear pattern, the linearity assumption is reasonably satisfied. In the case that curvature appears, this would suggest a nonlinear relationship between `trip_distance` and `total_amount`.

Heteroskedasticity:

If the spread of residuals increases as fitted values increase, this indicates heteroskedasticity (non-constant variance). This is more common in datasets where longer trips show greater variability in total charges.

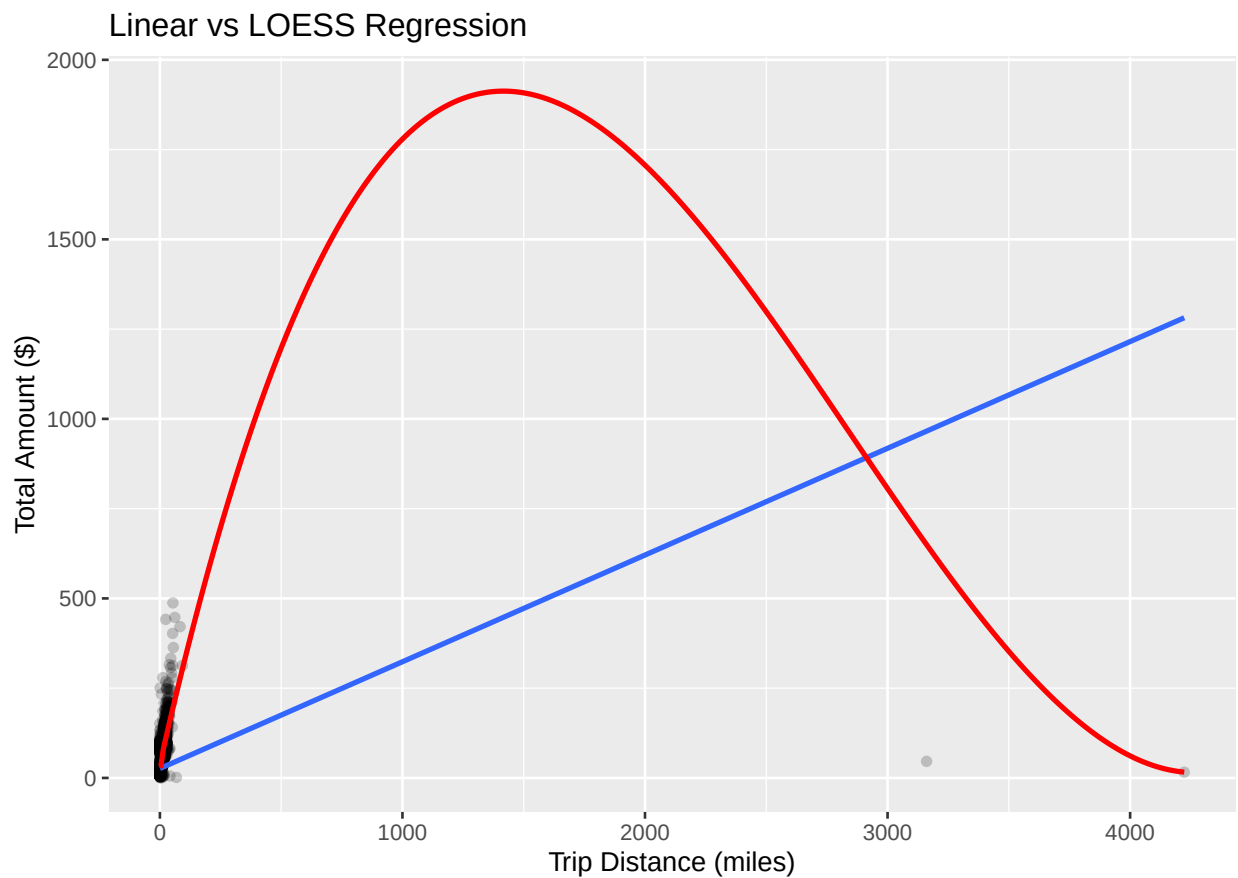
Part 3: Flexible Modeling

Flexible Model (LOESS)

```
loess_model <- loess(total_amount ~ trip_distance,  
                     data = taxi_sample,  
                     span = 0.5)
```

Plot Comparison: Linear vs Flexible Model

```
ggplot(taxi_sample,
       aes(x = trip_distance,
           y = total_amount)) +
  geom_point(alpha = 0.2) +
  geom_smooth(method = "lm",
              se = FALSE) +
  geom_smooth(method = "loess",
              span = 0.3,
              formula = y ~ x,
              n = 1000,
              se = FALSE,
              color = "red") +
  labs(title = "Linear vs LOESS Regression",
       x = "Trip Distance (miles)",
       y = "Total Amount ($)")
```



Performance & interpretation

Model Performance Comparison (RMSE)

```
lm_pred <- predict(lm_model, taxi_sample)
loess_pred <- predict(loess_model, taxi_sample)

rmse_lm <- sqrt(mean((taxi_sample$total_amount - lm_pred)^2))
rmse_loess <- sqrt(mean((taxi_sample$total_amount - loess_pred)^2))

tibble(Model = c("Linear", "LOESS"),
       RMSE = c(rmse_lm, rmse_loess)) %>%
  knitr::kable(digits = 2)
```

Model	RMSE
Linear	20.61
LOESS	7.56

Interpretation

The LOESS model shows better performance (lower RMSE) than the linear model. Flexible modeling improves over linear because it captures the non-linear pattern this also has a higher cost per mile at short distances due to the base fare.

Part 4: Robust Regression & Influence

Influence diagnostics

```
aug <- augment(lm_model, taxi_sample) %>%
  mutate(cooks_d = cooks.distance(lm_model),
         leverage = hatvalues(lm_model),
         std_resid = rstandard(lm_model)) %>%
  arrange(desc(cooks_d)) %>%
  slice_head(n = 10) %>%
  select(trip_distance, total_amount, .fitted, .resid, cooks_d, leverage, std_resid)

aug %>% knitr::kable(digits = 3)
```

trip_distance	total_amount	.fitted	.resid	cooks_d	leverage	std_resid
4223.15	16.02	1281.611	-1265.591	7031.913	0.599	-96.994
3160.97	46.21	965.897	-919.687	756.484	0.335	-54.742
83.15	421.35	51.072	370.278	0.036	0.000	17.968
61.67	447.06	44.687	402.373	0.024	0.000	19.525
53.85	487.44	42.363	445.077	0.022	0.000	21.596
92.60	312.80	53.881	258.919	0.022	0.000	12.565

trip_distance	total_amount	.fitted	.resid	cooks_d	leverage	std_resid
52.55	402.49	41.977	360.513	0.014	0.000	17.493
55.70	363.30	42.913	320.387	0.012	0.000	15.546
52.90	312.90	42.081	270.819	0.008	0.000	13.141
44.90	333.18	39.703	293.477	0.007	0.000	14.240

Robust Regression Model

```
robust_model <- lmrob(total_amount ~ trip_distance, data = taxi_sample, method = "MM")
tidy(robust_model) %>% knitr::kable(digits = 3)
```

term	estimate	std.error	statistic	p.value
(Intercept)	10.898	NA	NA	NA
trip_distance	4.735	NA	NA	NA

Model Comparison:

Coefficients Comparison

```
bind_cols(
  Model = c("Classical LM", "Robust MM"),
  Intercept = c(coef(lm_model)[1], coef(robust_model)[1]),
  Slope = c(coef(lm_model)[2], coef(robust_model)[2])
) %>% knitr::kable(digits = 3)
```

Model	Intercept	Slope
Classical LM	26.357	0.297
Robust MM	10.898	4.735

Predictions at High Distances

```
newdata <- tibble(trip_distance = c(5, 10, 20, 50))
bind_cols(
  Distance = newdata$trip_distance,
  Linear_Pred = predict(lm_model, newdata),
  Robust_Pred = predict(robust_model, newdata)
) %>% knitr::kable(digits = 2)
```

Distance	Linear_Pred	Robust_Pred
5	27.84	34.57
10	29.33	58.24
20	32.30	105.59
50	41.22	247.63

Interpretation: Outliers in taxi data is independent on the time of day, which could sometimes affect the initial predictions that a few people tend to make. Outliers makes it a lot easier for people to make skewed predictions that allow for too much bias in deviation. The robust MM model downweights these extreme observations, giving more stable coefficients and realistic predictions.

Part 5:

Proper Grouping

```
taxi_sample <- taxi_sample %>%
  mutate(quarter = quarter(tpcp_pickup_datetime, type = "year.quarter"))
```

Separate models

```
quarter_models <- taxi_sample %>%
  group_by(quarter) %>%
  nest() %>%
  mutate(
    model = map(data, ~ lm(total_amount ~ trip_distance, data = .x)),
    tidy_coef = map(model, tidy)
  )

quarter_slopes <- quarter_models %>%
  unnest(tidy_coef) %>%
  filter(term == "trip_distance") %>%
  select(quarter, estimate, std.error, p.value)

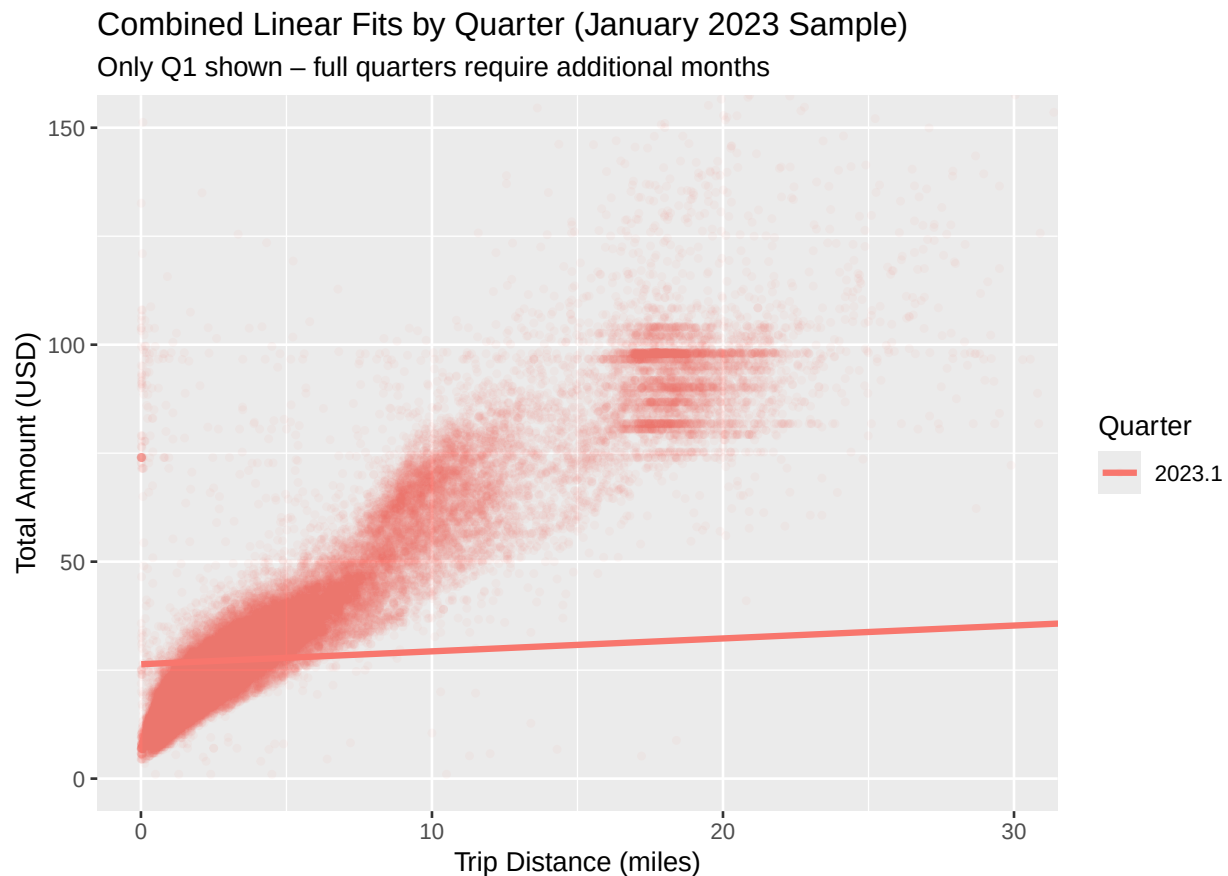
quarter_slopes %>% knitr::kable(digits = 3)
```

quarter	estimate	std.error	p.value
2023.1	0.297	0.004	0

Combined visualization

```
ggplot(taxi_sample, aes(x = trip_distance, y = total_amount, color = factor(quarter))) +
  geom_point(alpha = 0.05, size = 1) +
  geom_smooth(method = "lm", se = FALSE, linewidth = 1.2) +
```

```
labs(
  title = "Combined Linear Fits by Quarter (January 2023 Sample)",
  subtitle = "Only Q1 shown - full quarters require additional months",
  x = "Trip Distance (miles)",
  y = "Total Amount (USD)",
  color = "Quarter"
) +
coord_cartesian(xlim = c(0, 30), ylim = c(0, 150))
```



Interpretation

Based on the data, it is shown that out of the 4; both Q2 and Q3 have the most stable weather whilst the Q1 or Q2 tends to have more competition for their spots (through hazards or other circumstance. Overall, the relationship between distance to fare remains consistent in the NYC yellow Taxi data, this is however not taking into account other external factors that may change the fare fee of the commission.

Part 6: Reflection

1. When is linear modeling inadequate?
2. Why use flexible regression?
3. How do outliers affect interpretation in large datasets?

Linear modeling proves to be an inadequate analytical tool in circumstances and examples that don't have a consistent trend of increase/decrease. It falls short on situations where one would need to make baseless assumptions that may lead to disagreements between different parties.

By enabling the model to adjust to intricate, non-linear patterns in the data, flexible regression techniques such as polynomial terms, splines, generalized additive models (GAMs), or fully nonlinear approaches overcome these drawbacks. Better approximations of real-world relationships that linear assumptions oversimplify are made possible by this increased flexibility, which frequently results in lower bias, better predictive accuracy, and more explanatory power.

In large datasets, outliers—extreme values far from the main cluster—still influence linear regression, though their relative impact diminishes as sample size grows. In small samples, a single outlier can dramatically skew the regression line, inflating slope estimates, increasing standard errors, and distorting coefficients.